MILITARY SPECIFICATION

CASES, CARRYING, PHOTOGRAPHIC EQUIPMENT

This specification is approved for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for rigid, lightweight, hand carried, luggage cases for photographic equipment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-1	Standard For Laboratory Atmospheric Condition For Testing
FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection by Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6760

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SPECIFICATIONS

FEDERAL

TT-E-529	Enamel, Alkyd, Semi-gloss
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-W-8604	Welding, Of Aluminum Alloys, Process For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-W-52574	Welding Process And Welding Procedure Requirements For Manufacture of Equipment Utilizing Steels

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.3 Conflict with referenced documents. Where requirements stated in this specification conflict with any requirements in a referenced specification, the requirements of this specification shall apply.

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2), the contractor shall furnish sample carrying cases for first article inspection and approval (see 4.5).

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be of good commercial quality entirely suitable for the purpose for which they are used. Materials shall be free from all defects and imperfections that might affect serviceability or appearance of the product.

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3.2.1 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to atmospheric conditions likely to be encountered in storage or service.

3.2.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

3.2.2 Welding. The welding of aluminum shall be in accordance with MIL-W-8604. The welding of steel shall be in accordance with MIL-W-52574.

3.2.3 Screw threads. Screw threads used in the construction of the carrying cases shall conform to the requirements of the National Bureau of Standards Handbook H-28.

3.2.4 Fungus resistance. Materials not nutrients for fungus growth shall be used to the maximum extent possible. All other materials shall be treated to resist the growth of fungi.

3.2.5 Protective treatment. Materials used in the construction of the carrying cases that are subject to deterioration when exposed to environmental conditions likely to occur during storage or service usage shall be protected against deterioration in such a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes of environmental conditions shall be avoided.

3.3 Construction. Carrying cases shall be constructed of durable lightweight material. Carrying cases shall be fitted with partitions or shock-resistant cushions to protect the photographic equipment. Carrying cases shall be constructed to withstand, with a minimum of maintenance and readjustment, long periods of continuous service. Each assembly and part of the carrying cases shall be of sufficient strength and rigidity to accomplish its respective function and to maintain smooth operation with proper fit and alignment during the life of the carrying cases. Carrying cases shall utilize readily removable and replaceable parts to facilitate maintenance, repair, and adjustment. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other.

3.4 Performance. Carrying cases shall be capable of repeated openings and closings without the aid of tools or implements.

3.5 Enviromental conditions. Carrying cases shall be capable of performing satisfactorily when subjected to the environmental conditions specified herein.

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3.5.1 Temperature. Carrying cases shall be capable of withstanding prolonged storage at temperatures ranging from minus 60°F to plus 160°F. Carrying cases shall function properly at temperatures ranging from 0°F to plus 120°F.

3.5.2 Humidity. There shall be no failure or impairment of function when the carrying cases are subjected to relative humidity up to 95 percent at 120°F.

3.5.3 Salt spray. There shall be no failure or impairment of function when the carrying cases are subjected to salt sea atmosphere encountered under service conditions in naval operations.

3.5.4 Dust. The carrying cases shall provide dustproof protection for the interior of the cases.

3.5.5 Leakage. The carrying cases shall provide waterproof protection for the interior of the cases.

3.5.6 Impact. The carrying cases shall provide protection for contained photographic equipment against impact encountered during service use. When the carrying cases are subjected to impact there shall be no failure or impairment of function.

3.6 Major components. Carrying cases shall consist of the following major components:

Case shells
Hinge
Closure
Panel
Interior fittings
Latches
Locks
Lid support
Handle
Gliders

3.6.1 Case shells. When specified by the procuring activity, (see 6.2), the material used for the fabrication of the case shells shall be aluminum alloy, glass fiber reinforced plastic or other material having equal physical and corrosion-resistant properties. The shells shall be of uniform thickness and shall have sufficient strength to meet all requirements, specified herein. When required to meet strength requirements, the case shells shall be embossed with ribs of the appropriate shape to stiffen the case shells. The corners and edges of the case shells shall be smoothly rounded to a radius of 1.25-inches plus or minus 0.0625-inch.

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3.6.2 Hinges. The top and bottom case shells shall be joined together by hinges. The hinges shall be made from corrosion-resistant material and shall have the physical properties required to meet all requirements specified herein.

3.6.3 Closure. Carrying cases shall be provided with a closure that forms a dustproof and waterproof seal. The closure may be an integral part of the case shell or may be attached to the case shell by means of adhesive, riveting, or welding. The closure shall utilize a gasket of pure gum rubber, neoprene, or equal. When the closure is attached to the case shell by means of an adhesive, no air bubbles or excessive adhesive shall remain in the case shell-to-closure joint. The adhesive shall be high strength, properly cured, and shall not deteriorate due to aging and the environmental conditions specified herein. The portions of the closure forming the dustproof and waterproof seal shall be parallel to each other at the parting edge within 0.0625-inch. When the closure latches are in the locked position the gasket shall not be compressed more than 25 percent. Necessary guides shall be provided for aligning the top case shell with the bottom case shell during closing of the cases to insure minimum shifting of the two case shells.

3.6.4 Latches. The carrying cases shall be fitted with spring-loaded latches or drawbolts having sufficient physical and corrosion-resistant properties to meet all requirements specified herein. The spring-loaded latches or drawbolts shall be of sufficient quantity and spacing to accurately draw the two case shells together to effect a dustproof and waterproof seal.

3.6.5 Locks. The carrying cases shall be provided with tumbler-type lock assemblies or key-type lock assemblies as specified by the procuring activity (see 6.2).

3.6.6 Lid support. The carrying cases shall be provided with a folding lid support to serve as a limiting stop, hold the lid in the open position, and protect the carrying case hinge from damage.

3.6.7 Handle. The carrying cases shall be provided with a non-spring-loaded handle. The handle shall be capable of lifting and supporting a maximum load of 50 pounds. The handle shall be attached to the case at the center of balance. The handle shall be smoothly contoured to fit the hand and shall fold down against the case when not in use.

3.6.8 Gliders. The carrying cases shall be furnished with four corrosion-resistant gliders. The gliders shall be positioned at each of the four corners of the carrying case. The gliders shall support the carrying case in an upright position when placed on a flat surface, such as, a table top. The gliders shall provide clearance to protect the case finish and case hinge from damage.

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3.6.9 Panel. When specified by the procuring activity, (see 6.2), the top case shell shall be separated from the lower case shell by a hinged panel. The hinged panel shall isolate the contents of the top case shell from the contents in the lower case shell. The panel shall be constructed of a non-metallic material. The carrying cases shall be fitted with a latch to hold the panel in place.

3.6.10 Interior fittings. When specified by the procuring activity, (see 6.2), carrying cases shall be provided with the following interior fittings:

3.6.10.1 Fitted cushions. Carrying cases shall be furnished with polyurethane foam cushions. The cushions shall fit snugly into the top and bottom case shells. The cushions shall be furnished with cut-out or molded cavities to provide firm support for each item of photographic equipment.

3.6.10.2 Partitions. Carrying cases shall be fitted with partitions, arranged to provide separate compartments for each item of photographic equipment. The surfaces of the partitions shall be lined or padded to protect the photographic equipment from damage.

3.6.10.3 Blank cushions. The top and bottom case shells shall be entirely filled with a blank cushion of polyurethane foam.

3.7 Dimensions. The dimensions of the carrying cases shall be as specified by the procuring activity. (See 6.2).

3.8 Weight. The weight of the carrying cases, excluding photographic equipment, shall not exceed 20 pounds.

3.9 Finish. All exterior surfaces of aluminum case shells shall be buffed to a satin finish and shall be given a Type II, Class 1, anodic coating in accordance with MIL-A-8625. When specified, (see 6.2), case shells shall be painted with a high temperature baked enamel conforming to TT-E-529, Class B.

3.9.1 Color. The color of the Type II, Class 1 anodic coating and the color of the enamel shall be in accordance with FED-STD-595 as specified by the procuring activity (see 6.2).

3.10 Marking Identification marking of the carrying cases shall conform to MIL-STD-130.

3.10.1 Stock numbers. Upon request of the contractor, the procuring activity shall furnish the stock number for the carrying cases.

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3.11 Workmanship. The carrying cases, including all parts and accessories shall be fabricated and finished in a workmanlike manner with particular attention to burrs, sharp edges, accuracy of dimensions, thoroughness of welding, anodizing, painting, alignment of parts, and tightness of assembly screws, bolts, etc.

3.11.1 Dimensions and tolerances. Where dimensions and tolerances are not specified, and where they may affect the interchangeability, operation, or performance of the carrying cases, the dimensions and tolerances shall be held or limited accordingly.

3.11.2 Riveting. Riveting operations shall be performed to insure that rivets are tight and properly headed.

3.11.3 Screw assemblies. Assembly screws, bolts, and nuts shall be tightened and secured so as not to become loose due to vibration. The manner in which the screw assemblies are secured shall not prevent their removal for maintenance or repair of the carrying cases.

3.11.4 Preparatory cleaning. Prior to the application of any finish process, all surfaces shall be cleaned free from dirt, grease, oil, solderflux, visible rust, scale or oxides, and all other debris that might interfere with the adherence of the finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
b. Quality conformance inspection (see 4.6).

4.3 Test conditions.

4.3.1 Temperature and humidity. Unless otherwise specified herein, the carrying cases shall be tested under conditions of temperature and humidity as specified in FED-STD-No. 1.

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4.4 Failure during tests. Any characteristics exhibited by the carrying cases such as deterioration, binding, sticking, corrosion, etc., during any test specified herein which would cause the carrying case to fail to meet any of the requirements of Section 3 of this specification shall be the basis for rejection of the carrying cases.

4.5 First article inspection. The inspection shall consist of all inspections/tests specified herein.

4.5.1 First article sample. The first article sample shall consist of carrying cases that are representative models of the production carrying cases.

4.6 Quality conformance inspection. This inspection shall consist of the following inspections:

a. Individual tests (see 4.6.1).
b. Sampling tests (see 4.6.2).

4.6.1 Individual tests. Each carrying case shall be subjected to the following tests:

4.6.1.1 Physical examination. Examine each carrying case visually and with gauges to determine conformance to specified requirements for materials, construction, dimensions, weight, finish, color, marking, workmanship and other visible defects. Examine each carrying case to determine that the interior fittings conform to the specified requirements. Verify the proper operation of the hinges, locks, latches, lid support, handles, gliders, and interior panel. Examine each carrying case for fit and interchangeability of component parts.

4.6.2 Sampling tests. The sampling tests shall be in addition to the individual tests specified herein. Sampling for these tests shall be in accordance with MIL-STD-105 general inspection level II, utilizing the single sampling plan. The Acceptable Quality Level (AQL) shall be specified by the procuring activity (see 6.2). The lot size shall consist of all carrying cases offered for delivery at one time on any one contract or order.

4.6.2.1 Performance test. The carrying cases closure shall be subjected to 1,000 openings and 1,000 closings to determine conformance to 3.4 of this specification. Failure to pass this test shall be cause for rejection.

4.6.2.2 High temperature. The carrying cases shall be tested in accordance with Method 501.1, Procedure I of MIL-STD-810 to determine conformance to the requirements of 3.5.1 of this specification.

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4.6.2.3 Low temperature. The carrying cases shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810, except that the minimum temperature shall be minus 60°F, to determine conformance to the requirements of 3.5.1 of this specification.

4.6.2.4 Humidity. The carrying cases shall be tested in accordance with Method 507.1, Procedure V of MIL-STD-810 to determine conformance to the requirements of 3.5.2 of this specification.

4.6.2.5 Fungus. The carrying cases shall be tested in accordance with Method 508.1, Procedure I of MIL-STD-810. Upon completion of the test, the presence of any fungus growth which caused deterioration in any material contained in the case assembly or would impair the functions of the carrying case shall be cause for rejection.

4.6.2.6 Salt spray. The carrying cases shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810 to determine conformance to the requirements of 3.5.3 of this specification.

4.6.2.7 Dust. The carrying cases shall be tested in accordance with Method 510.1, Procedure I of MIL-STD-810 to determine conformance to the requirements of 3.5.4 of this specification.

4.6.2.8 Rain. The carrying cases shall be tested in accordance with Method 506.1, Procedure I of MIL-STD-810 to determine conformance to the requirements of 3.5.5 of this specification.

4.6.2.9 Impact. The carrying cases shall be subjected to 6 drops onto a concrete surface from a height of 36 inches. The case shall be held by the handle for each drop. The case shall contain the cushioning material and photographic equipment as specified by the procuring activity (see 6.2). At the conclusion of the test there shall be no damage to the cases or component parts which would impair the intended functions. There shall be no damage to the contained photographic equipment.

4.7 Rejection and retest. Rejected carrying cases shall be replaced or reworked to correct the defects, after which all necessary tests shall be repeated. Rejected carrying cases shall not be resubmitted for inspection without furnishing complete particulars concerning previous rejections and the measures taken to correct the defects. When investigation indicates that the defects may exist in carrying cases previously accepted, complete particulars concerning the defects found, including recommendations for correction, shall be furnished to the procuring activity.

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4.8 Destroyed specimen. All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.9 Examination of packaging and packing. Examine the packaging and packing of the carrying cases in accordance with the Quality Assurance Provisions of the specifications referenced in Section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified by the procuring activity (see 6.2).

5.1.1 Level A. Each carrying case shall be individually packaged in accordance with Method III of MIL-P-116. The container shall be a snug-fitting fiberboard box conforming to Type CF, Class Weather Resistant, Variety SW, Grade W6C, Style RSC or OSC of PPP-B-636. Boxes shall be inspected, closed, sealed, and strapped in accordance with the applicable provisions of PPP-B-636. (See 6.2).

5.1.2 Level C. Carrying cases shall be packaged in accordance with the manufacturer's commercial practice, and in a manner to afford protection against corrosion, deterioration, and physical damage during direct shipment from the supplier to the user. (See 6.2).

5.2 Packing. Packing shall be level A or C as specified. (See 6.2).

5.2.1 Level A. Carrying cases preserved and packaged level A, as specified herein, shall be packed in accordance with the applicable provisions of MIL-STD-794. (See 6.2).

5.2.2 Level C. Carrying cases preserved and packaged level C, as specified herein, shall be packed for shipment to insure carrier acceptance and safe delivery at the destination. (See 6.2).

5.3 Marking. The interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The carrying cases covered in this specification are intended to provide complete protection for photographic equipment during service use.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity of carrying cases required.
c. Quantity of first article samples required. (See 3.1).
d. Type of material used for case shells. (See 3.6.1).
e. Type of lock required. (See 3.6.5).
f. Panel, if required. (See 3.6.9).
g. Type of interior fittings required. (See 3.6.10).
h. Carrying case dimensions. (See 3.7).
i. Painting required. (See 3.9).
j. Color of carrying case. (See 3.9.1).
k. Acceptable quality level. (See 4.6.2).
l. Type of cushioning material and type of photographic equipment required for drop test. (See 4.6.2.9).
m. Level of preservation, packaging, and packing required. (See Section 5).

Custodians:

Army - EL
Navy - AS
Air Force - 99

Preparing Activity:

Navy - AS

(Project No. 6760-0552)

Review Activities:

Army - EL
Defense - GS

User Activities:

Army - EL
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